

MASENO UNIVERSITY

MAS 812: NON-PARAMETRIC METHODS ASSIGNMENT

Introduction to non-parametric methods

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Single-sample and paired-sample methods

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EXERCISE 9.1

1. Given the sample 12, 15, 10, 18, 20

a) Compute the order statistics

Step I: Order the data as

10, 12, 15, 18, 20 and $n = 5$

Step II: Compute order statistics [range, Quartiles, median and IQR]

Range: $20 - 10 = 10$

Median: $X_{\frac{5+1}{2}} = X_3 = 15$

Quartiles: upper quartile, $Q_3 = 18$, lower quartile, $Q_1 = 12$

IQR: $Q_3 - Q_1 = 18 - 12 = 6$

b) Construct the empirical distribution function

$$F_n(x) = \begin{cases} 0, & x < 10 \\ \frac{1}{5}, & 10 \leq x < 12, \\ \frac{2}{5}, & 12 \leq x < 15, \\ \frac{3}{5}, & 15 \leq x < 18, \\ \frac{4}{5}, & 18 \leq x < 20, \\ 1, & x \geq 20 \end{cases}$$

c) Find the sample median

Given $n = 5$, sample median is given by $X_{\frac{5+1}{2}} = X_3 = 15$

EXERCISE 9.2

Let $X_1, \dots, X_n$ be i.i.d from $F(x)$. Show that $E[F_n(x)] = F(x)$. Derive $Var[F_n(x)]$

EXERCISE 9.3

2. For $n = 10$ and $F(x_0) = 0.3$, compute

a. $E[F_n(x)]$

$E[\hat{F}_n(x_0)] = F(x_0) = 0.3$

b. $Var[F_n(x)]$
 $Var[F(x_0)] = \frac{0.3(1 - 0.3)}{10} = 0.021$

c. The approximate standard error
 $sd = \sqrt{0.021} \approx 0.1449$

EXERCISE 9.4

Discuss situations where a nonparametric test would be preferred over a parametric one

1. *When the sample size is small*
 Parametric tests rely on large-sample properties or assumptions about the population distribution. Small sample sizes make it difficult to verify these assumptions.
2. *When the data is ordinal*
 Parametric tests are conducted on ratio scale data. However, this changes when the data in question possesses an ordinal/ranking nature. This is where non-parametric techniques become of use. Examples include likert scale survey responses and literal rankings made on observations.
3. *When data is highly skewed*
 Parametric tests are sensitive to extreme values, because they rely on the mean and standard deviation. If the dataset contains strong outliers, they may distort results. Non-parametric tests are more robust because they typically use medians or ranks instead of means.

Single-sample and paired-sample methods

1. A company claims that the median battery life of their new product is 10 hours. A consumer group tests 15 batteries and obtains the following lifetimes (hours):

9.8, 10.2, 9.9, 10.5, 9.7, 10.1, 9.6, 10.3, 9.8, 10.4, 9.9, 10.0, 9.5, 10.2, 9.8

- a. Perform a sign test at $\alpha = 0.05$ to test if the median differs from 10 hours

```
> battery_test <- binom.test(x = sum(battery > 10),
+                             n = sum(battery != 10),
+                             p = 0.5,
+                             alternative = "two.sided", conf.level = 0.95);cat(paste
("p-value:",battery_test$p.value))
p-value: 0.79052734375
```

- b. Construct a 95% confidence interval for the median using the sign test.

```
> battery_95ci <- SignTest(x = battery,
+                           alternative = "two.sided",
+                           mu = 10, conf.level = 0.95);cat(paste("95
% confidence bounds: [",battery_95ci$conf.int[1],",",battery_95ci$co
nf.int[2],"]"))
95% confidence bounds: [ 9.8 , 10.2 ]
```

- c. Interpret your results

The median battery life of the new product is 10 hours at a significance level of 5%. Further, we are 95% confident that the median battery life in hours lies between 9.8 and 10.2 hours.

2. Twelve patients are given a drug to lower blood pressure. Systolic blood pressure is measured before and after treatment:

patient	1	2	3	4	5	6	7	8	9	10	11	12
before	145	138	152	140	148	135	150	142	146	138	144	140
after	138	132	144	138	142	130	142	136	140	132	138	134

- a. Compute the differences (Before - After).

```
> bp$difference <- bp$before-bp$after;bp$difference
[1] 7 6 8 2 6 5 8 6 6 6 6 6
```

Perform a Wilcoxon signed-rank test to determine if the drug reduces blood pressure.

- b. Use $\alpha = 0.05$.

```
> bp_test <- wilcox.test(x = bp$difference,
+                         alternative = "greater",
+                         mu = 0, conf.level = 0.95,
+                         exact = TRUE);bp_test$p.value
```

Warning message:

```
In wilcox.test.default(x = bp$difference, alternative = "greater", :
  cannot compute exact p-value with ties
[1] 0.001005306
```

c. Compute the Hodges-Lehmann estimator of the median reduction.

```
> bp_hl <- HodgesLehmann(x = bp$before, y = bp$after);bp_hl
[1] 6
```

d. Construct a 95% confidence interval for the median reduction.

```
> bp_95ci <- wilcox.test(x = bp$difference,
+                         mu = 0,
+                         conf.int = TRUE, conf.level = 0.95);bp_95ci
$conf.int
[1] 5.499986 6.999984
attr(,"conf.level")
[1] 0.95
```

e. Compare your results with a paired t-test. Which do you trust more and why?

```
> bp_test2 <- t.test(x = bp$before,
+                    y = bp$after,
+                    alternative = "greater",
+                    paired = TRUE,
+                    mu = 0,
+                    conf.level = 0.95);bp_test2$p.value
[1] 2.450014e-09
```

```
> Kurt(bp$difference) # Kurtosis
[1] 1.32569
```

```
> Skew(bp$difference) # Skewness
[1] -1.100751
```

Both tests return a small p-value in comparison to 0.05, in favor of the alternative hypothesis of significant blood pressure change.

The Wilcoxon test is never much worse than the t-test (minimum ARE = 0.864), but can be dramatically better for heavy-tailed distributions such as the one encountered in the above example.

3. Generate data from the following distributions
in R ($n = 20$ each)

- a. Normal(0,1)
- b. Cauchy (location=0, scale=1)
- c. Exponential(1) shifted to have median 0

For each data;

- Test $H_0: \theta = 0$ vs $H_1: \theta \neq 0$ using t-test, sign test, and Wilcoxon test.
- Repeat 1000 times and compute the empirical Type I error rate (should be near 0.05)
- Now shift the distributions by adding 0.5 to each observation and compute power
- Summarize your findings in a table

What do you conclude about the robustness and power of these methods?

```
# I. Data generation
> set.seed(123)
> norm <- rnorm(n = 20, mean = 0, sd = sqrt(1))
> cau <- rcauchy(n = 20, location = 0, scale = 1)
> expo <- rexp(n = 20, rate = 1) - log(2)

## Table of p-value for the 3 tests and distributions
      t_test sign_test Wilcoxon_sign_rank_test
Normal    0.11704653 0.1153183             0.11398697
Cauchy     0.70544307 1.0000000             0.78412628
Exponential 0.05258354 0.5034447             0.04405403

# 8 of the 9 hypotheses tested suggest a failure to reject the null hypothesis
of a median value of 0. The only exception came from a Wilcoxon sign rank test
conducted on randomly exponentially shifted values whose decision was to
reject the null hypothesis in favor of the claim of a non-zero median.

# II. Type I error rate
### Table of type I error rates from the 1000 simulations
      t_test sign_test Wilcoxon_sign_rank_test
Normal    0.050      0.039             0.047
Cauchy     0.023      0.039             0.047
Exponential 0.190      0.039             0.047

# i. t-test: Type one error rate is highest in the exponentially distribute
d values at 19%. The Cauchy values record the lower rate at 2.3%
# ii. Sign test: the error rate is constant across all 3 distributed values
at 3.9%
# iii. Wilcoxon sign rank test: Similar to the sign test, the type I error
rate is constant across the 3 distributions at 4.7%

# III. Statistical power
### Table of statistical powers
      t_test sign_test Wilcoxon_sign_rank_test
Normal    0.575      0.399             0.555
Cauchy     0.592      0.401             0.558
Exponential 0.578      0.410             0.555
```

```
# The table above shows statistical power for 3 tests conducted on
Normal, Cauchy and Exponential observations, randomly simulated 1000
times and increased by 0.5.
# On average, the sign test possesses the least statistical power across
all 3 distributions compared to the t and Wilcoxon sign rank test.
# t-test and Wilcoxon sign rank rank possess higher power than the sign
test but the Wilcoxon sign rank test has more consistent power values.
# Across all 3 distributions, the t-test returned the highest power with
the Cauchy distributed values returning the highest at 59.2%
```

4. Consider the dataset:

3.2, 4.1, 2.8, 3.9, 4.5, 3.7, 2.9, 4.2, 3.5, 3.8

a. Compute the sample mean and median

```
> x_bar <- mean(x, na.rm = TRUE); m_bar <- median(x, na.rm = TRUE);
cat(paste("sample mean:", x_bar, "\n", "sample median:", m_bar))
sample mean: 3.66
sample median: 3.75
```

b. Compute all Walsh averages and find the Hodges-Lehmann estimator

```
> ## Walsh averages and the Hodges-Lehmann estimator
> w <- outer(X = x, Y = x, FUN = function(a,b){(a+b)/2}) # repeated
Walsh averages
> w_values <- w[upper.tri(x, diag = TRUE)] |> sort(); w_values # Uniq
ue and ordered Walsh averages_
[1] 3.00 3.05 3.20 3.35 3.45 3.50 3.55 3.65 3.70 3.85
> x_hl <- HodgesLehmann(x = w_values); cat(paste("Hodges-Lehmann est
imator:", x_hl)) # Hodges-Lehmann estimator
Hodges-Lehmann estimator: 3.45
```

c. Add an outlier of 20 to the dataset and recompute all three estimators

```
> x2 <- append(x, 20)
> x2_bar <- mean(x2, na.rm = TRUE); m2_bar <- median(x2, na.rm = T)
> w2 <- outer(X = x2, Y = x2, FUN = function(a,b){(a+b)/2})
> w2_values <- w2[upper.tri(x2, diag = TRUE)] |> sort()
x2_hl <- HodgesLehmann(x = w2_values)
> cat(paste("sample mean:", x2_bar, "\n", "sample median:", m2_bar, "\n
", "Hodges-Lehmann estimator:", x2_hl))
sample mean: 5.14545454545455
sample median: 3.8
Hodges-Lehmann estimator: 3.5
```

d. Discuss the robustness of each estimator

	first	second	difference
sample mean	3.66	5.145455	1.485455
sample median	3.75	3.800000	0.050000
Hodges-Lehmann estimator	3.45	3.500000	0.050000

The Hodges-Lehmann estimator and the sample median experience the least change when the outlier value is added to the initial dataset. On the other hand the sample mean exhibits the largest change when the new value is added. This shows that both the Hodges-Lehmann estimator and the sample median are significantly more robust to outliers as compared to the sample mean.

5. An agricultural experiment tests two varieties of wheat. Ten plots are each split into two halves, with one variety planted in each half

plot	1	2	3	4	5	6	7	8	9	10
Variety A	42	38	45	40	43	41	39	44	42	40
Variety B	40	37	44	39	42	40	38	43	41	38

- a. Are these paired or independent samples? Justify.

Paired. Both varieties of wheat are planted on each of the same plot. Had both varieties been planted on 10 different plots, the samples would be independent.

- b. Test whether variety A yields more than variety B using an appropriate non-parametric test

```
> agric_test <- wilcox.test(x = agric$a, y = agric$b,
+                           alternative = "greater",
+                           paired = TRUE, exact = TRUE);cat("p-value:",pas
te(agric_test$p.value))

p-value: 0.0017622499059455
```

- c. Estimate the median difference in yield

```
> agric_sample_median <- median(x = (agric$a-agric$b), na.rm = TRUE)
;agric_sample_median
[1] 1
```

- d. Provide a 90% confidence interval for the median difference

```
> agric_90ci <- wilcox.test(x = agric$a, y = agric$b,
+                           paired = TRUE, exact = TRUE,
+                           conf.int = TRUE, conf.level = 0.90);cat(
paste("90% confidence bounds:[",agric_90ci$conf.int[1],",",agric_90c
i$conf.int[2],"]")
90% confidence bounds: [ 1 , 1.00005824043857 ]
```

e. Write a brief conclusion for an Agricultural researcher

```
# Wheat variety A produces more yield than variety B at a significance
level of 5%, given the same condition of the plots. Moreover, we are 90%
confident that variety A produces between 1 and 1.00005824043857 more
yield than wheat variety B given the same condition of the plots.
```

6. Conceptual questions (answer briefly)

a. Why does the sign test ignore zeros? What would happen if we included them as "positive" or "negative"?

The sign-test is based on categorizing observations in a dichotomous manner with respect to the null hypothesis. Identifying zeros as either positives or negatives would introduce bias to the sign-test and hence threaten the credibility of its results.

b. Explain the symmetry assumption required for the Wilcoxon signed-rank test. Why is this important?

The Wilcoxon-signed rank test uses both sign and magnitude. Here, absolute paired differences are computed and ranked. Finally, the ranks are classified as either positives or negatives with respect to the claimed median value under the null hypothesis.

c. Under what conditions would you prefer the sign test over the Wilcoxon test?

- *When the distribution of the paired differences violates the WSR test assumption of symmetry*
- *Dealing with ordinal data with unclear spacing (likert-scale)*
- *When we have multiple tied differences.*

d. What does the Hodges-Lehmann estimator estimate when the distribution is not symmetric?

*The Hodges-Lehmann estimator estimates the **pseudomedian** which is a median of all pairwise averages of independent observations from the distribution*

e. How would you handle tied observations in the Wilcoxon signed-rank test?

When two or more ranks are equal in magnitude, we replace them by their mid-rank